

Cybersecurity Risk Assessment & Security Improvement Project

Emerald Financial Solutions Ltd
Prepared by: Thenuki Munasinghe
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*****Note:** Emerald Financial Solutions Ltd is a fictional company created for this project. All information and data used are fictional and are for educational and portfolio purposes only.

1. Introduction

This project was created to carry out a cybersecurity risk assessment for Emerald Financial Solutions Ltd, a fictional financial services company based in Dublin.

The main aim of the project was to look at the company's current IT environment, identify its important assets, identify possible cybersecurity threats and assess the level of risk these threats could cause.

After identifying the risks, I looked at ways the company could improve its security. This included creating security recommendations, policies and an implementation roadmap.

I also created an Excel dashboard to give a simple visual overview of the company's assets, risks, risk levels, budget allocation and implementation progress.

2. Project Purpose

The main purpose of this project was to understand how a real organisation could identify and manage cybersecurity risks.

The project focused on questions such as:

- What are the company's most important assets?
- What threats could affect these assets?
- How serious are the different risks?
- Which risks should be dealt with first?
- What security controls could reduce these risks?
- What security policies should employees follow?
- How could the recommended improvements be implemented?
- How can the company's cybersecurity progress be monitored?

This project also allowed me to develop practical skills in cybersecurity, risk assessment, documentation, Excel and project planning.

3. Company Overview

Emerald Financial Solutions Ltd is a fictional financial services company based in Dublin.

The company was founded in 2018 and has approximately 45 employees and around 2,500 customers.

The company operates using a hybrid working model, with employees working three days in the office and two days remotely.

Company Details

Category	Details
Company	Emerald Financial Solutions Ltd
Industry	Financial Services
Location	Dublin, Ireland
Founded	2018
Employees	45
Customers	Approximately 2,500
Revenue	Approximately €7.5 million
Working Model	Hybrid
Business Hours	Monday–Friday, 9:00–17:30

4. Departments

The company has the following departments:

- Management – 4 employees
- Finance – 8 employees
- Customer Support – 10 employees
- Sales – 12 employees
- HR – 3 employees
- IT – 5 employees
- Compliance – 3 employees

Different departments handle different types of company and customer information, which means they can also face different cybersecurity risks.

5. Information Handled

The company handles a range of important and sensitive information, including:

- Customer names
- Customer addresses
- Phone numbers
- Bank account details
- Mortgage documents
- Payroll information
- HR records
- Emails

Bank account details and mortgage documents were treated as particularly important because a security breach involving this information could have a serious impact on both customers and the company.

6. Project Objectives

The main objectives of the project were to:

1. Identify the company's important assets.
 2. Identify potential cybersecurity threats.
 3. Assess the likelihood and impact of each threat.
 4. Identify the highest-priority risks.
 5. Recommend appropriate security improvements.
 6. Create security policies for employees.
 7. Create an implementation roadmap.
 8. Create a professional cybersecurity dashboard.
 9. Improve the company's overall security position.
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7. IT Environment

The company uses a number of different technologies and systems, including:

- Windows 11 laptops
- Desktop PCs
- Microsoft 365
- Microsoft Teams
- SharePoint
- VPN
- Wi-Fi
- File server

- Network switches
- Firewall
- NAS backup
- Printers
- Company phones
- CCTV

I also created a network diagram showing how the main components of the company's network are connected.

The diagram includes the internet, ISP, firewall, router, managed switch, office computers, printers, Wi-Fi, server, NAS backup and Microsoft 365.

8. Physical Security

I also looked at the physical security of the company's office.

The office was divided into three main security zones:

Zone 1 – Public: Reception

Zone 2 – Staff: Open-plan office

Zone 3 – Restricted: IT office and server room

Some physical security controls were already in place, including CCTV and staff access cards.

However, I identified areas that could be improved, including visitor escorting and the introduction of a clean desk policy.

Some of the physical risks identified included:

- Unauthorised people entering the office
- Laptop theft
- Tailgating
- Unauthorised server room access
- Visitors entering restricted areas
- Theft of confidential documents
- Fire damage
- Water damage
- Unlocked computers
- Loss of backup devices

9. Asset Register

I created an asset register to identify the main technology assets used by the company.

Some of the key assets identified were:

Asset	Quantity	Risk Classification
Employee Laptops	45	High
Desktop PCs	5	High
Network Switches	2	Critical
Firewall	1	Critical
Server	1	Critical
NAS Backup	1	Critical
Wi-Fi Access Points	3	High
Printers	3	Medium
Company Phones	45	Medium
CCTV Cameras	4	Medium
Backup Drives	2	High

The asset register helped me understand which assets would need the greatest level of protection.

10. Threat Model

After identifying the assets, I created a threat model to identify the main threats that could affect the company.

The ten main threats identified were:

1. Phishing Attack
2. Ransomware
3. Weak Passwords
4. Business Email Compromise
5. Malware Infection
6. Laptop Theft
7. Unauthorised Network Access
8. Unauthorised Physical Access
9. VPN Account Compromise
10. Insider Threat

These threats were selected because they could have a significant effect on the company's systems, information and employees.

11. Risk Assessment

I used a **5 × 5 risk matrix** to assess the identified threats.

Each threat was given a:

- Likelihood score
- Impact score

The risk score was then calculated using:

Risk Score = Likelihood × Impact

The score was used to determine the overall risk level.

Risk Assessment Results

Threat ID	Threat	Likelihood	Impact	Score	Risk Level
THR-001	Phishing Attack	5	5	25	Critical
THR-002	Ransomware	3	5	15	High
THR-003	Weak Passwords	5	4	20	Critical
THR-004	Business Email Compromise	3	5	15	High
THR-005	Malware Infection	3	4	12	High
THR-006	Laptop Theft	3	4	12	High
THR-007	Unauthorised Network Access	3	5	15	High
THR-008	Unauthorised Physical Access	3	4	12	High
THR-009	VPN Account Compromise	3	4	12	High
THR-010	Insider Threat	2	5	10	Medium

12. Main Risk Findings

The risk assessment showed that **Phishing Attack** was the highest-rated threat, with a score of **25 (Critical)**.

Phishing was considered a major risk because employees could potentially be tricked into giving away their login details or clicking on malicious links or attachments.

The second Critical risk was **Weak Passwords**, which received a score of **20**.

Other important High risks included:

- Ransomware
- Business Email Compromise
- Malware
- Laptop Theft
- Unauthorised Network Access
- Unauthorised Physical Access
- VPN Account Compromise

These risks should be prioritised when implementing security improvements.

13. Gap Analysis

I created a gap analysis to identify areas where the company's current security could be improved.

The main areas identified included:

- Password security
- Multi-factor authentication
- Employee cybersecurity awareness
- Email security
- Endpoint protection
- Backup and recovery
- Network security
- Physical security
- Remote working security
- Visitor management
- Clean desk practices
- Security monitoring

The gap analysis helped me decide which improvements should be included in the recommendations and implementation plan.

14. Security Recommendations

Based on the risks and gaps identified, I developed a number of recommendations.

Multi-Factor Authentication

MFA should be used for important accounts, particularly Microsoft 365 and VPN accounts.

Stronger Passwords

The company should introduce and enforce a strong password policy.

Employee Training

Employees should receive regular cybersecurity awareness training, especially around phishing and social engineering.

Email Security

Additional email security controls should be used to reduce phishing and business email compromise.

Endpoint Protection

Company laptops and computers should have appropriate protection against malware and ransomware.

Backup Improvements

Backups should be protected and regularly tested to make sure data can be recovered if required.

Network Security

The firewall, VPN and network access controls should be regularly reviewed.

Physical Security

Access to restricted areas should be controlled and visitors should be properly managed.

15. Security Policies

As part of the project, I created six separate security policies:

- **Password Policy**
- **Remote Working Policy**
- **Email Security Policy**
- **Incident Reporting Policy**
- **USB Device Policy**
- **Bring Your Own Device (BYOD) Policy**

These policies are designed to give employees clear instructions on how they should handle cybersecurity risks in their everyday work.

16. Risk Treatment

The main approach used to deal with the identified risks was to **reduce the risk** by introducing appropriate security controls.

The Critical risks should be addressed first, followed by the High and Medium risks.

Examples of risk treatment include:

- Using MFA to reduce account compromise.
- Providing security awareness training to reduce phishing risk.
- Improving endpoint protection to reduce malware and ransomware risk.
- Improving backups to support recovery.
- Strengthening physical access controls.
- Improving VPN security.
- Introducing clear security policies.

17. Implementation Roadmap

I created an implementation roadmap to show how the recommended improvements could be introduced.

Phase 1 – Critical Risks

The first phase focuses on the most serious risks, including:

- Phishing
- Weak passwords
- Account security
- MFA
- Email security

Phase 2 – High Risks

The second phase focuses on:

- Ransomware
- Malware
- Network security
- VPN security
- Backup improvements

Phase 3 – Physical and Policy Improvements

This phase focuses on:

- Physical access
- Visitor management
- Clean desk practices
- Remote working
- USB security
- BYOD

Phase 4 – Ongoing Improvement

The final phase focuses on:

- Monitoring
- KPI tracking
- Regular risk reviews
- Policy reviews
- Backup testing
- Security awareness

18. Security Budget

A separate security budget document was not created as part of the current project.

The dashboard does include a **budget allocation** section. However, I would not include made-up costs in this report.

A future version of the project could include a detailed budget covering:

- Security software
- MFA
- Endpoint protection
- Employee training
- Backup solutions
- Email security
- Network security
- Physical security
- Implementation costs

The costs should be based on realistic supplier or licence prices before being used for an actual business decision.

19. Professional Dashboard

I created a professional Excel dashboard to provide a quick overview of the cybersecurity assessment.

The dashboard includes charts showing:

- Number of assets
- Number of risks
- Risk levels
- Budget allocation
- Implementation progress

The dashboard makes it easier to understand the main results of the project without having to go through every individual spreadsheet.

It could also be useful for management because it provides a simple way of monitoring cybersecurity progress.

20. KPIs

The project can be monitored using several Key Performance Indicators (KPIs).

Some suitable KPIs include:

Critical Risks Remaining

The number of Critical risks that still need to be addressed.

High Risks Remaining

The number of High risks that remain open.

Implementation Progress

The percentage of planned security actions that have been completed.

Security Policy Completion

The percentage of required security policies that have been implemented.

Staff Security Training

The percentage of employees who have completed cybersecurity awareness training.

Backup Testing

The percentage of planned backup and recovery tests that have been successfully completed.

These KPIs can be reviewed regularly to see whether the organisation's security is improving.

21. Monitoring and Review

Cybersecurity should not be treated as a one-time task.

The company should regularly:

- Review its risk register.
- Reassess important risks.
- Review security policies.
- Monitor Critical and High risks.
- Test backups.
- Review employee training.
- Monitor implementation progress.
- Review cybersecurity KPIs.
- Update the threat model when new threats appear.

Regular reviews would help ensure that the security controls remain effective as the company and its technology change.

22. Conclusion

Overall, this project allowed me to carry out a complete cybersecurity risk assessment for a fictional financial services company.

I started by looking at the company and identifying its important assets. I then created a threat model and assessed the main cybersecurity risks using likelihood and impact.

The assessment identified **Phishing Attack** and **Weak Passwords** as the two Critical risks, with several other threats being classified as High.

I then developed security recommendations, a gap analysis, security policies and an implementation roadmap to show how these risks could be reduced.

Finally, I created an Excel dashboard to present the main results visually.

The project helped me develop practical skills in **cybersecurity risk assessment, threat modelling, risk analysis, asset management, security policy development, Excel, project planning and cybersecurity documentation.**

It also gave me experience of looking at cybersecurity from a more practical and business-focused point of view rather than only focusing on technical solutions.

23. Project Deliverables

The completed project includes:

- Company Overview
 - Asset Register
 - Asset Summary
 - Network Diagram
 - Threat Model
 - Risk Assessment
 - Risk Matrix
 - Gap Analysis
 - Security Recommendations
 - Password Policy
 - Remote Working Policy
 - Email Security Policy
 - Incident Reporting Policy
 - USB Device Policy
 - BYOD Policy
 - Implementation Roadmap
 - Professional Dashboard
 - Final Report
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24. Project Skills Demonstrated

Through this project, I developed and demonstrated skills in:

- Cybersecurity Risk Assessment
- Threat Modelling
- Risk Analysis
- Risk Matrix Development
- Asset Management
- Gap Analysis
- Security Recommendations
- Security Policy Development
- Project Planning
- Excel
- Dashboard Development
- Documentation
- IT Project Management
- Cybersecurity Awareness

Final Portfolio Note

Emerald Financial Solutions Ltd is a fictional organisation created specifically for this project. All company details, assets, risks and other information have been created for educational and portfolio purposes. No confidential or real customer information has been used.